Appendix D - Per-Engine MMIO Maps

Every memory-mapped device, grouped by subsystem. Each register is 32-bit on the bus unless noted; only the documented low bits carry information.

D.1 Video chip (`$F0000`-`$F049B`)

Address	Name	R/W	Notes
`$F0000`	`VIDEO_CTRL`	R/W	Bit `0` enable; bit `1` completed-frame presentation hold.
`$F0004`	`VIDEO_MODE`	R/W	Mode value selector.
`$F0008`	`VIDEO_STATUS`	R	Bit `0` `HAS_CONTENT`, bit `1` `VBLANK`, bit `2` `FB_ERR`.
`$F000C`	`COPPER_CTRL`	R/W	Copper enable.
`$F0010`	`COPPER_PTR`	R/W	Copper-list base address.
`$F0014`	`COPPER_PC`	R	Current copper program counter.
`$F0018`	`COPPER_STATUS`	R	Copper running / stopped.
`$F001C`	`BLT_CTRL`	R/W	Blitter control.
`$F0020`	`BLT_OP`	R/W	Blitter operation.
`$F0024`	`BLT_SRC`	R/W	Source address.
`$F0028`	`BLT_DST`	R/W	Destination address.
`$F002C`	`BLT_WIDTH`	R/W	Width, byte count for `MEMCOPY`, or packed endpoint.
`$F0030`	`BLT_HEIGHT`	R/W	Height or packed endpoint.
`$F0034`	`BLT_SRC_STRIDE`	R/W	Source stride.
`$F0038`	`BLT_DST_STRIDE`	R/W	Destination stride.
`$F003C`	`BLT_COLOR`	R/W	Fill colour or scale destination size.
`$F0040`	`BLT_MASK`	R/W	Mask address.
`$F0044`	`BLT_STATUS`	R	Blitter busy / done.
`$F0048-$F0054`	`VIDEO_RASTER_*`	R/W	Raster split lines and per-line palette swaps.
`$F0058-$F0074`	`BLT_MODE7_*`	R/W	Mode 7 affine texture registers.
`$F0078`	`VIDEO_PAL_INDEX`	R/W	CLUT palette write index.
`$F007C`	`VIDEO_PAL_DATA`	R/W	CLUT palette data.
`$F0080`	`VIDEO_COLOR_MODE`	R/W	`0` RGBA32, `1` CLUT8.
`$F0084`	`VIDEO_FB_BASE`	R/W	Framebuffer base address.
`$F0088-$F0487`	palette	R/W	256-entry direct palette table.
`$F0488-$F049B`	`BLT_EXT_*`	R/W	`BLT_FLAGS`, `BLT_FG`, `BLT_BG`, `BLT_MASK_MOD`, `BLT_MASK_SRCX`.

`VIDEO_MODE` values:

Value	Frame size
`$00`	`640` by `480`
`$01`	`800` by `600`
`$02`	`1024` by `768`
`$03`	`1280` by `960`
`$04`	`320` by `200`
`$05`	`320` by `240`
`$06`	`1920` by `1080`
`$07`	`960` by `540`

`BLT_OP` values:

Value	Operation
`0`	`COPY`
`1`	`FILL`
`2`	`LINE`
`3`	`MASKED_COPY`
`4`	`ALPHA_COPY`
`5`	`MODE7`
`6`	`COLOR_EXPAND`
`7`	`SCALE`
`8`	`MEMCOPY`

For `MEMCOPY`, `BLT_SRC` is the source byte address, `BLT_DST` is the destination byte address, and `BLT_WIDTH` is the byte count.

`BLT_FLAGS` bits `0`-`1` select RGBA32 or CLUT8 where the operation supports it, bits `4`-`7` select the raster operation, bit `11` selects MSB-first mask sampling for `MASKED_COPY`, and bit `12` selects alpha-template source bytes for `ALPHA_COPY`.

D.2 Terminal / serial / input (`$F0700`-`$F07FF`)

Address	Name	R/W	Notes
`$F0700`	`TERM_OUT`	W	Print one byte.
`$F0704`	`TERM_STATUS`	R	Bit `0` input avail, bit `1` output ready.
`$F0708`	`TERM_IN`	R	Dequeue cooked input byte.
`$F070C`	`TERM_LINE_STATUS`	R	Bit `0` complete line in queue.
`$F0710`	`TERM_ECHO`	R/W	Local echo.
`$F0724`	`TERM_CTRL`	R/W	Bit `0` line-input mode.

Address	Name	R/W	Notes
$F0728	TERM_KEY_IN	R	Dequeue raw key.
$F072C	TERM_KEY_STATUS	R	Bit 0 raw key avail.
$F0730	MOUSE_X	R	Absolute X (low 16 bits).
$F0734	MOUSE_Y	R	Absolute Y.
$F0738	MOUSE_BUTTONS	R	Bit 0 L, 1 R, 2 M.
$F073C	MOUSE_STATUS	R	Bit 0 changed (clears on read).
$F0740	SCAN_CODE	R	Dequeue raw scancode.
$F0744	SCAN_STATUS	R	Bit 0 scancode avail.
$F0748	SCAN_MODIFIERS	R	Bit 0 shift, 1 ctrl, 2 alt, 3 caps.
$F074C	MOUSE_CTRL	R/W	Bit 0 request relative mode.
$F0750	RTC_EPOCH	R	Seconds since 1970-01-01 00:00:00 UTC.
$F0754	MOUSE_DX	R	Signed accumulated dX (clears on read).
$F0758	MOUSE_DY	R	Signed accumulated dY.
$F075C	RTC_MONO_USEC_LO	R	Low 32 bits of monotonic microseconds since engine start.
$F0760	RTC_MONO_USEC_HI	R	High 32 bits of monotonic microseconds since engine start.
$F07F0	TERM_SENTINEL	W	Write $DEAD to stop CPU.

D.3 SoundChip ($F0800-$F0B7F)

Range	Block	Notes
$F0800-$F08FF	Global	AUDIO_CTRL, ENV_SHAPE, filter, reverb, overdrive.
$F0900-$F093F	Square	SQ_FREQ, SQ_DUTY, SQ_ENV_*, SQ_GATE.
$F0940-$F097F	Triangle	Same shape.
$F0980-$F09BF	Sine	Same shape.
$F09C0-$F09FF	Noise	LFSR period, volume, gate.
$F0A00-$F0A1F	Sync / ring-mod	Source-channel selects.
$F0A20-$F0A6F	Sawtooth	Same shape.
$F0A80-$F0B7F	Flex 0-3	64-byte per-channel block.

Audio player control blocks in the same area:

Range	Block	Registers
$F0B80-$F0B91	AHX	AHX_PLUS_CTRL, AHX_PLAY_PTR, AHX_PLAY_LEN, AHX_PLAY_CTRL, AHX_PLAY_STATUS, AHX_SUBSONG.
$F0BA0-$F0BBF	MIDI/MUS	MIDI_PLAY_PTR, MIDI_PLAY_LEN, MIDI_PLAY_CTRL, MIDI_PLAY_STATUS (bit 0 busy, bit 1 error, bit 2 paused, bit 3 loading), MIDI_POSITION, MIDI_VOLUME, MIDI_TEMPO_BPM.

Range	Block	Registers
`$F0BC0-$F0BD7`	MOD	`MOD_PLAY_PTR`, `MOD_PLAY_LEN`, `MOD_PLAY_CTRL`, `MOD_PLAY_STATUS`, `MOD_FILTER_MODEL`, `MOD_POSITION`.
`$F0BD8-$F0BF3`	WAV	`WAV_PLAY_PTR`, `WAV_PLAY_LEN`, `WAV_PLAY_CTRL`, `WAV_PLAY_STATUS`, `WAV_POSITION`, `WAV_PLAY_PTR_HI`, `WAV_CHANNEL_BASE`, `WAV_VOLUME_L`, `WAV_VOLUME_R`, `WAV_FLAGS`.
`$F0BF4-$F0BF6`	Live MIDI	`IE_MIDI_LIVE_DATA` byte write stream, `IE_MIDI_LIVE_STATUS` bit `0` active, `IE_MIDI_LIVE_CTRL` bit `0` reset. Data and control writes are port writes, not RAM shadow bytes.

D.4 SFX triggers

Per-channel offset	Field	Notes
`+$00`	`SFX_PTR`	Sample base address.
`+$04`	`SFX_LEN`	Sample length in bytes.
`+$08`	`SFX_LOOP_PTR`	Loop point.
`+$0C`	`SFX_LOOP_LEN`	Loop length.
`+$10`	`SFX_FREQ`	Playback rate.
`+$14`	`SFX_VOL`	Volume field; audible level is clipped to `0-255`.
`+$16`	`SFX_PAN_RESERVED`	Reserved pan field.
`+$18`	`SFX_FORMAT`	Sample format.
`+$1C`	`SFX_CTRL`	Trigger / one-shot / loop.

Extended window: 32 channels at `$F2600-$F29FF`, stride `$20`. Legacy aliases: channels `0-3` at `$F0E80`, `$F0EA0`, `$F0EC0`, and `$F0EE0`.

D.5 PSG / AY-3-8910 (`$F0C00-$F0C20`)

Offset	Register
`+$00-+$0F`	R0-R15, one byte per register (byte-contiguous).
`+$10`	`PSG_PLAY_PTR` (32-bit).
`+$14`	`PSG_PLAY_LEN` (32-bit).
`+$18`	`PSG_PLAY_CTRL` (bit `0` start, bit `1` stop, bit `2` loop).
`+$1C`	`PSG_PLAY_STATUS` (bit `0` busy, bit `1` error).
`+$20`	`PSG_PLUS_CTRL`.

D.6 SN76489 (`$F0C30-$F0C3F`)

Offset	Register
`+$00`	`SN_PORT_WRITE`.
`+$01`	`SN_PORT_READY`.

Offset	Register
+$02	SN_PORT_MODE.

D.7 SID family

Each SID instance is a byte-contiguous block in the original 6581/8580 register layout. Three independent instances live at three bases:

Block	Range	Notes
SID	$F0E00-$F0E1C	Primary SID.
SID2	$F0E30-$F0E4C	Second independent SID.
SID3	$F0E50-$F0E6C	Third independent SID.

Within each instance the offsets follow the original chip layout, one byte per register:

Offset	Register
+$00-+$06	Voice 1: FREQ_LO, FREQ_HI, PW_LO, PW_HI, CTRL, AD, SR.
+$07-+$0D	Voice 2 (same shape).
+$0E-+$14	Voice 3 (same shape).
+$15-+$17	Filter: FC_LO, FC_HI, RES_FILT.
+$18	MODE_VOL.
+$19	SID+ control (SID_PLUS_CTRL) / POT_X mirror.
+$1A	POT_Y.
+$1B	OSC3.
+$1C	ENV3.

The primary SID block at $F0E00 has an attached SID file player block after the voice registers:

Offset (from $F0E00)	Register
+$20	SID_PLAY_PTR (32-bit).
+$24	SID_PLAY_LEN (32-bit).
+$28	SID_PLAY_CTRL (start / stop / loop).
+$2C	SID_PLAY_STATUS (busy / error).
+$2D	SID_SUBSONG.

The two flexible audio blocks $F0C40-$F0CFF and $F0D40-$F0DFF are not SID2 and SID3. They are extra-channel mirrors (channels 4-6 and 7-9 in the global SoundChip channel space) and have the flex-channel layout documented under SoundChip (D.3), not the SID layout. A program that wants SID2 / SID3 should program them at $F0E30 and $F0E50.

D.8 POKEY (`$F0D00`-`$F0D20`)

Offset	Register
+$00	POKEY_AUDF1.
+$01	POKEY_AUDC1.
+$02	POKEY_AUDF2.
+$03	POKEY_AUDC2.
+$04	POKEY_AUDF3.
+$05	POKEY_AUDC3.
+$06	POKEY_AUDF4.
+$07	POKEY_AUDC4.
+$08	POKEY_AUDCTL.
+$09	POKEY_PLUS_CTRL.
+$0A	POKEY_RANDOM (read).
+$10	SAP_PLAY_PTR.
+$14	SAP_PLAY_LEN.
+$18	SAP_PLAY_CTRL.
+$1C	SAP_PLAY_STATUS.
+$20	SAP_SUBSONG.

D.9 TED audio (`$F0F00`-`$F0F1F`)

Offset	Register
+$00	TED_FREQ1_LO.
+$01	TED_FREQ2_LO.
+$02	TED_FREQ2_HI.
+$03	TED_SND_CTRL.
+$04	TED_FREQ1_HI.
+$05	TED_PLUS_CTRL.
+$10	TED_PLAY_PTR.
+$14	TED_PLAY_LEN.
+$18	TED_PLAY_CTRL.
+$1C	TED_PLAY_STATUS.

D.10 TED video (`$F0F20`-`$F0F6B`)

40 by 25 text with 121-colour cells, raster position registers, raster compare registers, and raster pending status; see Chapter 6.

D.11 VGA (`$F1000`-`$F13FF`)

Range	Subsystem
`$F1000`-`$F1008`	`VGA_MODE`, `VGA_STATUS`, `VGA_CTRL`.
`$F1010`-`$F1018`	Sequencer (`VGA_SEQ_*`).
`$F1020`-`$F102C`	CRTC (`VGA_CRTC_*`).
`$F1030`-`$F103C`	Graphics controller (`VGA_GC_*`).
`$F1040`-`$F1044`	Attribute controller (`VGA_ATTR_*`); mode bit 3 selects blink or bright-background use of text attribute bit 7.
`$F1050`-`$F105C`	DAC + palette index/data.
`$F1100`-`$F13FF`	Palette RAM: 256 contiguous three-byte RGB entries, six bits stored per component.

In text mode, the CRTC start and cursor addresses count cells in the 16-K-cell text aperture. Display addressing wraps modulo 16384 cells while preserving each character and attribute pair.

D.12 ULA (`$F2000`-`$F2017`)

Offset	Register
`+$00`	`ULA_BORDER` (bits 0-2).
`+$04`	`ULA_CTRL`.
`+$08`	`ULA_STATUS`.
`+$0C`	`ULA_ADDR_LO`.
`+$10`	`ULA_ADDR_HI`.
`+$14`	`ULA_DATA`.

ULA VRAM aperture at `$FA000`-`$FBAFF`.

D.13 ANTIC + GTIA (`$F2100`-`$F21FB`)

ANTIC block at `$F2100`-`$F213F` with `DMACTL`, `CHACTL`, `DLISTL/H`, `HSCROL`, `VSCROL`, `PMBASE`, `CHBASE`, `WSYNC`, `VCOUNT`, `NMIEN/IST/RES`. GTIA at `$F2140`-`$F21FB` with `COLPM0-3`, `COLPF0-3`, `COLBK`, `PRIOR`, `GRACTL`, `CONSOL`, player/missile positions, sizes, graphics bytes, collision latches, and `HITCLR`.

D.14 File I/O (`$F2200`-`$F221F`, IE64 extension `$F22B0`)

Used by `LOAD`, `SAVE`, `BLOAD`, `COMPILE`, `TRANSPILE`, `ASSEMBLE`, `DIR`, and `TYPE`, and by machine code that talks to the disk volume directly.

Offset	Register
+$00	`FILE_NAME_PTR`.
+$04	`FILE_DATA_PTR`.
+$08	`FILE_DATA_LEN` (write byte count; ignored by read).
+$0C	`FILE_CTRL` (write triggers).
+$10	`FILE_STATUS`.
+$14	`FILE_RESULT_LEN` (actual read/list byte count; `0` after accepted-path read failure).
+$18	`FILE_ERROR_CODE`.
+$1C	`FILE_READ_MAX` (one-shot read cap in bytes; larger file refused with `FILE_ERR_RANGE` before any byte is copied; `0` unbounded).
$F22B0	`FILE_DATA_PTR64` (IE64-only `64`-bit read/write data-buffer pointer; extension, not a replacement for `FILE_DATA_PTR`).

`FILE_ERROR_CODE` values:

Value	Meaning
0	OK.
1	Not found.
2	Permission.
3	Path traversal.
4	Range error: staged data would reach `$FFFF0000`, wrap the `32`-bit pointer, or exceed active RAM.

D.15 Amiga Paula DMA (`$F2260`-`$F22AF`)

Four-channel DMA sample engine. Each channel is `16` bytes:

Offset	Register
+$00	`PTR` sample pointer.
+$04	`LEN` length in words.
+$08	`PER` period.
+$0C	`VOL` volume.

Global registers:

Address	Register
$F22A0	`AROS_AUD_DMACON`.
$F22A4	`AROS_AUD_STATUS`.
$F22A8	`AROS_AUD_INTENA`.

D.16 Media loader (`$F2300`-`$F231F`)

Dispatches by file extension into the matching audio engine. Used by `SOUND PLAY` (Chapter 23).

Offset	Register
+$00	MEDIA_NAME_PTR.
+$04	MEDIA_SUBSONG.
+$08	MEDIA_CTRL (1 play, 2 stop).
+$0C	MEDIA_STATUS (0 idle, 1 loading, 2 playing, 3 error).
+$10	MEDIA_TYPE (1 SID, 2 PSG, 3 TED, 4 AHX, 5 POKEY, 6 MOD, 7 WAV, 8 MIDI/MUS).
+$14	MEDIA_ERROR (0 OK, 1 not found, 2 bad format, 3 unsupported, 4 name invalid, 5 too large).

D.17 RUN loader block ($F2320-$F233F)

The RUN keyword uses this block for service launches. It is not a normal reader programming route; use RUN syntax, COCALL, or IE Mon as described in the main chapters.

Offset	Field
+$00	name pointer.
+$04	control.
+$08	status.
+$0C	type.
+$10	error.
+$14	session.

D.18 Coprocessor ($F2340-$F238F)

COSTART, COSTOP, COCALL, COSTATUS, and COWAIT use this command block:

Offset	Register
+$00	COPROC_CMD.
+$04	COPROC_CPU_TYPE.
+$08	COPROC_CMD_STATUS.
+$0C	COPROC_CMD_ERROR.
+$10	COPROC_TICKET.
+$14	COPROC_TICKET_STATUS.
+$18	COPROC_OP.
+$1C	COPROC_REQ_PTR.
+$20	COPROC_REQ_LEN.
+$24	COPROC_RESP_PTR.
+$28	COPROC_RESP_CAP.
+$2C	COPROC_TIMEOUT.

Offset	Register
+$30	COPROC_NAME_PTR.
+$34	COPROC_WORKER_STATE.
+$38	COPROC_STATS_OPS.
+$3C	COPROC_STATS_BYTES.
+$40	COPROC_IRQ_CTRL.
+$44	COPROC_DISPATCH_OVERHEAD.
+$48	COPROC_COMPLETED_TICKET.
+$4C	COPROC_INSTANCE.

Command values written to COPROC_CMD:

Value	Command
1	COPROC_CMD_START, start worker from a named file.
2	COPROC_CMD_STOP, stop the selected worker.
3	COPROC_CMD_ENQUEUE, submit a request and return a ticket.
4	COPROC_CMD_POLL, poll the ticket in COPROC_TICKET.
5	COPROC_CMD_WAIT, wait for the ticket in COPROC_TICKET.
6	COPROC_CMD_START_MEM, start worker from the guest-RAM image at COPROC_REQ_PTR for COPROC_REQ_LEN bytes.

Extended monitor block ($F23B0-$F23BF):

Offset	Register
+$00	COPROC_RING_DEPTH.
+$04	COPROC_WORKER_UPTIME.
+$08	COPROC_STATS_RESET, write 1 to clear operation and byte counters and restart busy accounting.
+$0C	COPROC_BUSY_PCT, rolling worker busy percentage over about one second.

COPROC_INSTANCE selects the worker instance (0 = default) alongside COPROC_CPU_TYPE. M68K, x86, and IE64 accept instances 0 and 1; IE32, 6502, and Z80 accept instance 0 only.

Capability and version-discovery block ($F25A0-$F25BF), sited in the free gap between the CPU Wait block and the SFX extended aliases:

Offset	Register
+$00	COPROC_INSTANCE_LIMIT, instance count for the selected COPROC_CPU_TYPE.
+$04	COPROC_SELECTED_STATE, running state of the selected (type, instance).
+$08	COPROC_MAILBOX_VERSION, mailbox layout version.
+$0C	COPROC_WORKER_BASE, selected worker window base address.
+$10	COPROC_WORKER_END, selected worker window end address.
+$14	COPROC_WORKER_RING, selected worker ring base address.

Offset	Register
+$18	`COPROC_INSTANCE_STATE`, per-`(cpuType, instance)` running bitmask (bit `cpuType*2 + instance`). One atomic read; needs no selector write.

The coprocessor mailbox occupies `$790000-$792FFF`: twelve ring slots at a `$400` stride, ring index `cpuTypeIndex * 2 + instance`. Each ring stores head, tail, capacity, layout version, and worker acknowledgement at offsets `+$00` through `+$04`; requests begin at `+$08` and responses at `+$208`.

D.19 IRQ diagnostics (`$F23C0`-`$F23DF`)

Offset	Register
+$00	In-service mask.
+$04	State flags.
+$08	Pending mask.
+$0C	Delivered counters.
+$10	Blocked counters.
+$14	RTE count.
+$18	Consecutive STOP spins.
+$1C	Watchdog event count.

D.20 SysInfo (`$F2400`-`$F24FF`)

Offset	Register
+$00	`SYSINFO_TOTAL_RAM_LO`.
+$04	`SYSINFO_TOTAL_RAM_HI`.
+$08	`SYSINFO_ACTIVE_RAM_LO`.
+$0C	`SYSINFO_ACTIVE_RAM_HI`.
+$10	`SYSINFO_FEATURES`: bit `0` CPU Wait, bit `1` Voodoo command stream, bit `2` MMIO stats enabled, bit `3` Voodoo texture slots.
+$14	`SYSINFO_LOW_WINDOW_LO`.
+$18	`SYSINFO_LOW_WINDOW_HI`.

D.21 Shared network sockets (`$F2500`-`$F257F`)

Nine 32-bit registers point at one fixed 96-byte big-endian request descriptor in guest RAM.

Offset	Register	Notes
+$00	`HOST_SOCKET_CMD`	Write command `1` through `23` to dispatch.
+$04	`HOST_SOCKET_REQ_PTR`	Guest request-descriptor address.
+$08	`HOST_SOCKET_REQ_LEN`	Descriptor byte length; use `96`.

Offset	Register	Notes
+$0C	HOST_SOCKET_RES1	Primary result or $FFFFFFFF.
+$10	HOST_SOCKET_RES2	Secondary result.
+$14	HOST_SOCKET_ERRNO	Socket error.
+$18	HOST_SOCKET_HERRNO	Resolver error.
+$1C	HOST_SOCKET_STATUS	0 ready, 1 busy, 2 error.
+$20	HOST_SOCKET_EVENTS	Pending event bits.

Commands are SOCKET, BIND, LISTEN, ACCEPT, CONNECT, SENDTO, RECVFROM, SHUTDOWN, SETSOCKOPT, GETSOCKOPT, GETSOCKNAME, GETPEERNAME, IOCTL, CLOSE, WAITSELECT, GETHOSTBYNAME, GETHOSTBYADDR, GETHOSTNAME, DUP2, GETEVENTS, RELEASE, RELEASECOPY, and OBTAIN, numbered 1 through 23 in that order.

IE64, IE32, M68K, and x86 use the physical addresses. The 6502 and Z80 select Bank 3 value $79 and use $6500 through $657F. Their byte lanes are big-endian, and a byte-written command dispatches only after all four command lanes have been written. The block is not mapped in BASIC mode. See Chapter 39.

D.22 HOST appliance block ($F1400-$F140F)

Offset	Name	Width	Notes
+$00	command	byte	Subverb enum (W).
+$04	trigger	byte	Non-zero fires the command (W).
+$08	status	byte	Terminal-state enum (R).
+$0C	exit	32-bit	Full 32-bit exit code from the underlying action (R); byte lanes at +$0C-+$0F return successive bytes of the same value.

See Chapter 36 for the subverb enum and the state machine. Reachable from IE64, IE32, M68K, and x86; not reachable from the 6502 or Z80.

D.23 CPU Wait ($F2580-$F259F)

Offset	Register
+$00	CPU_WAIT_VBLANK: write parks until the next VBlank rising edge, waiting out any current VBlank first.
+$04	CPU_WAIT_UNTIL_LO: low 32 bits of the RTC_MONO_USEC deadline.
+$08	CPU_WAIT_UNTIL_HI: high 32 bits of the RTC_MONO_USEC deadline.
+$0C	CPU_WAIT_UNTIL_GO: write parks until monotonic time reaches the latched deadline.

Waits have a short safety timeout so a programme continues if video presentation is unavailable.

D.24 Gamepad input ($F25C0-$F25FF)

Updated once per displayed frame. Reads have no side effects. Writes are accepted but ignored, and disconnected slots read as zero.

Offset	Register
+$00	GAMEPAD_STATUS: bits 0..3 pad connected mask, bits 8..11 connected count.
+$04-+$0F	Reserved; reads return zero.

Each pad p (0..3) has a 12-byte record at $F25D0 + p*$0C:

Offset	Register
+$00	BUTTONS: canonical button bitfield, latched at frame start.
+$04	AXIS_LXY: left stick, X in low 16 bits, Y in high 16 bits (signed).
+$08	AXIS_RXY: right stick, X in low 16 bits, Y in high 16 bits (signed).

Button bits: 0 Up, 1 Down, 2 Left, 3 Right, 4 A, 5 B, 6 X, 7 Y, 8 LB, 9 RB, 10 LT, 11 RT, 12 Select, 13 Start, 14 L3, 15 R3, 16 Home. Triggers are digital. Axes range from -32767 through 32767; left and up are negative, right and down are positive. Documented in Chapter 37.

D.25 Voodoo 3D ($F8000-$F87FF)

Status, framebuffer base, clip rect, triangle setup, texture descriptors, fog, alpha, chroma-key, Z-buffer. Documented in Chapter 9. Texture RAM at $D0000-$DFFFF.

TRIANGLE_CMD queues a triangle and binds the current raster state and uploaded texture to that triangle. Later Voodoo register writes or texture uploads affect later triangles only. A full 4096-triangle batch is flushed into the drawing buffer without publishing, then submission continues. SWAP_BUFFER_CMD hands the current batch to the rasteriser and publishes the visible frame. Up to two swap jobs may be active; a later swap waits only when that pipeline is full. FBI_BUSY and SST_BUSY report render or publish work in progress; SWAPBUF reports a pending publish swap. MEMFIFO is a coarse current-batch ready field, and PCIFIFO is the high-level command-space field.

Command-stream registers:

Address	Register
$F833C	VOODOO_CMD_PTR, guest-RAM stream pointer.
$F8340	VOODOO_CMD_COUNT, number of address/value pairs.
$F8344	VOODOO_CMD_SUBMIT, write 1 for big-endian pairs or 2 for little-endian pairs.

Submit value VOODOO_CMD_SUBMIT_REPLAY (1) reads big-endian register addresses and values. VOODOO_CMD_SUBMIT_REPLAY_LE (2) reads both words as little-endian values. Replay uses the same register path as ordinary Voodoo writes. Misaligned addresses, out-of-range addresses, and writes back to the command-stream control registers are skipped. The maximum count is 65536 pairs.

Retained texture-slot registers:

Address	Register
$F8350	VOODOO_TEX_SLOT, identifier retained by the next texture upload.
$F8354	VOODOO_TEX_BIND, make a retained texture current without another texel transfer.

Slot identifiers 0 through 65535 are valid. Writing $FFFFFFFF to VOODOO_TEX_SLOT disables retention for later uploads. An invalid slot selection or a bind to an empty slot leaves the previous selection or current texture unchanged. Test bit 3 of SYSINFO_FEATURES before using this extension.